1 **(a)** 1 **[B1]**

(b) Ordered list: $0, 0, 1, 1, 1, 1, 2, 2, 3, 4$ **[M1]**

1 **[A1]**

(c) $\dfrac{2+0+1+4+1+0+3+1+2+1}{10} = \dfrac{15}{10}$ **[M1]**

1.5 **[A1]**

[Total Marks: 5]

2 **(a)** 16 **[B1]**

(b) 21 **[B1]**

(c) Ordered list seen: $5, 8, 10, 13, 14, 16, 18, 20, 22, 24, 26$ **[M1]**

12 **[A1]**

(d) $(14 + 20 + 8 + 24 + 16 + 5 + 26 + 13 + 22 + 10 + 18) \div 11$ or $176 \div 11$ **[M1]**

16 **[A1]**

[Total Marks: 6]

3 **(a)** 7 **[B1]**

(b) $12 - 7 = 5$ **[B1]**

(c) 8 **[B1]**

(d) $(7 + 7 + 7 + 8 + 10 + 12 + 12) \div 7$ or $\dfrac{63}{7}$ **[M1]**

9 **[A1]**

(e) $\dfrac{63 + y}{8} = 9.5$ or 9.5×8 **[M1]**

$76 - 63 = 13$ **[A1]**

[Total Marks: 7]

4 **(a)** $30 < h \leq 40$ **[B1]**

(b) Uses midpoints $15, 25, 35, 45, 55$ **[M1]**

$\Sigma fx = (15 \times 12) + (25 \times 25) + (35 \times 28) + (45 \times 11) + (55 \times 4)$ or
$180 + 625 + 980 + 495 + 220 = 2500$ **[M1]**

$\dfrac{2500}{80}$ **[M1]**

31.25 **[A1]**

[Total Marks: 5]

5 **(a)** $100 < m \leq 110$ **[B1]**

(b) Midpoints used: $85, 95, 105, 115, 125$ **[M1]**

$\Sigma fx = 85 \times 8 + 95 \times 22 + 105 \times 31 + 115 \times 14 + 125 \times 5$ **[M1]**

$\Sigma fx = 680 + 2090 + 3255 + 1610 + 625 = 8260$ **[M1]**

$\text{Mean} = \dfrac{8260}{80} = 103.25$ **[A1]**

(c) The exact masses of individual apples are not known **[B1]**

The calculation uses the midpoints of the class intervals, assuming the data is evenly distributed within each class **[B1]**

[Total Marks: 7]

6 (a) $30 < t \leq 40$ **[B1]**

(b) Midpoints: $15, 25, 35, 45, 55$ **[M1]**

$\Sigma fx = 15(14) + 25(28) + 35(32) + 45(18) + 55(8)$ **[M1]**

$= 210 + 700 + 1120 + 810 + 440 = 3280$ **[M1]**

$\dfrac{3280}{100} = 32.8$ **[A1]**

(c) $18 + 8 = 26$ **[M1]**

26 **[A1]**

(d) $40 < t \leq 50$ **[B1]**

[Total Marks: 8]

7 (a) $p + q = 25$ **[B1]**

$\dfrac{10p + 420 + 800 + 70q + 450}{60} = 48$ or $10p + 70q = 1210$ [B2] (Allow B1 for unsimplified $\Sigma fx = 10p + 420 + 800 + 70q + 450$ seen)

Correctly sets up and solves simultaneous equations to eliminate one variable **[M1]**

$p = 9$, $q = 16$ **[A1]**

(b) Identifies median is between 30th and 31st value and shows cumulative frequencies 23 and 39 (or $9 + 14 = 23$ and $23 + 16 = 39$) **[M1]**

$40 < m \leq 60$ **[A1]**

[Total Marks: 7]

8 (a) Since the mode is 8, there must be at least two 8s. **[M1]**

Since the median is 11, the middle number when ordered is 11. The first three numbers are $8, 8, 11$. **[M1]**

The mean is 12, so the sum of the five numbers is $5 \times 12 = 60$. **[M1]**

The range is 15, so the largest number is $8 + 15 = 23$. **[M1]**

Let the fourth number be x. Then $8 + 8 + 11 + x + 23 = 60 \implies 50 + x = 60 \implies x = 10$. **[A1]**

(b) New sum $= 6 \times 13 = 78$ **[M1]**

$k = 78 - 60 = 18$ **[A1]**

[Total Marks: 7]

9 **(a)** $15 + x + y + 8 + 2 = 50$ leading to $x + y = 25$ **[B1]**

(b) Midpoints: $1, 3, 5, 7, 9$ **[M1]**

$$\frac{1(15) + 3(x) + 5(y) + 7(8) + 9(2)}{50} = 3.8$$ **[M1]**

$15 + 3x + 5y + 56 + 18 = 190$ **[M1]**

$3x + 5y + 89 = 190$ leading to $3x + 5y = 101$ **[A1]**

(c) $3x + 3y = 75$ or substitution method $3(25 - y) + 5y = 101$ **[M1]**

$2y = 26 \implies y = 13$ **[A1]**

$x = 12$ **[A1]**

(d) $2 < m \leq 4$ **[B1]**

[Total Marks: 9]